You'll add a new unique constraint that defines uniqueness on both City and State.

A UNIQUE constraint is added using an ALTER TABLE statement with the ADD CONSTRAINT keyword. PostgreSQL automatically adds an index when you add a constraint, to allow a fast lookup when enforcing the constraint.

Since PostgreSQL adds an index anyway, adding a Unique Index is a common way to add a Unique constraint. For a running system, adding the constraint via the Unique index makes it possible to add the index CONCURRENTLY and avoid a long exclusive lock of the table where the index is being added. This is a good design goal because a long-held lock might block other queries trying to write and read to the table.

If you're a Rails developer you've added Unique indexes before, so you can skip this refresher section. If you're newer to Rails, the conventional way to add a Unique index would be to use the CONCURRENTLY keyword and add these changes to a new Active Record Migration.

From your terminal, generate a new Migration by running bin/rails generate migration and give it a name. Below, "g" is short for "generate". Generation in Rideshare may fail because this Migration exists already.

```
bin/rails g migration AddLocationsCityStateUnique
```

Open the generated file and fill in the add_index method, setting the unique option and concurrently options as follows. This migration was added to Rideshare earlier.

```
class AddLocationsCityStateUnique < ActiveRecord::Migration[7.1]
  disable_ddl_transaction!

  def change
    add_index :locations, [:city, :state],
      unique: true,
      algorithm: :concurrently
  end
end
```

The migration was applied running bin/rails db:migrate. UNIQUE constraints are added smoothly when there aren't existing violations and there are multiple rows with the same combination of city and state.

Often when redefining uniqueness things don't go so smoothy!

When rows with duplicate values exist, PostgreSQL won't allow the constraint to be added. Thanks to Transactional DDL, the Constraint creation DDL rolls